Crypto Basis Mean-Reversion Strategy
Strategy Design Document

Terence Chiu
tchiu04@uw.edu
University of Washington
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Strategy Overview

This document describes a statistical arbitrage strategy that exploits the persistent basis relationship between cryptocurrency perpetual futures and spot prices. The core hypothesis is that the basis—defined as the normalized spread between the perpetual (perp) price and the spot price—is mean-reverting, and that funding rate regimes identified via a Hidden Markov Model (HMM) can serve as a sentiment oscillator to improve entry timing and reduce adverse selection.

The strategy trades on the BTC/BTC-USDH and GOLD-USDH pairs, entering positions when the market is in a "crowd long" regime (positive funding) and the basis is sufficiently elevated, then exiting when the regime shifts or volatility becomes extreme.

Strategy Testing, Constraints, and Objectives

Testing Methodology

Backtesting is conducted using both spot and perpetual futures data with the following parameters:

- Walk-forward backtests on spot and perp data.
- 8-month rolling training windows with 1-month walk-forward evaluation periods.
- Various lookback windows for rolling mean and volatility estimation.
- Performance evaluation across Information Ratio, Calmar Ratio, Sharpe Ratio, turnover, drawdown, and robustness metrics.

Constraints

- Leverage caps, liquidity filters, and position limits are applied.
- Benchmarks include BTC buy-and-hold and a market-cap-weighted crypto index.

Risk Management

- **Exit the trade when:** the funding rate switches sides.
- **Adjust position when:** volatility is forecasted to be high.

Validation

Statistical validation of the spread is performed using ADF and Phillips-Perron (PP) tests on both the raw basis and the demeaned basis (Perp − Spot).

Building Blocks

Filter

Instruments traded: BTC/BTC-USDH, GOLD-USDH.

Indicators

Spread Indicators

- **Basis:** Defined as $\text{Basis}_t = (\text{Perp}_t - \text{Spot}_t)/\text{Spot}_t$.
- **Basis Moving Average (z-score):** Rolling standardized deviation of the basis from its mean.
- **Basis Change (Velocity):** First difference of the basis, capturing rate of change.
- **NOFI:** Net Order Flow Imbalance, measuring directional pressure in the order book.

Regime Indicator

- **HMM Regime Probabilities:** A Hidden Markov Model is fitted to funding rate features to classify the market into distinct sentiment regimes.

Risk Indicators

- **Realized Volatility:** 24-hour rolling annualized volatility.
- **Volatility Ratio:** Ratio of short-term to long-term realized volatility, used to detect volatility regime shifts.

Funding Indicators

- **Funding Rate:** The periodic payment between long and short perpetual holders.
- **Funding Rate Moving Average (8-hour):** Smoothed funding rate used to assess structural drift in sentiment.

Additional indicators considered but not yet incorporated: Perp Open Interest, Kalman Filtered velocity.

Signals

Three primary signals are constructed:

S1: Distance from Local Mean A z-score signal derived from the basis moving average. Returns +1 if the z-score exceeds the entry threshold (basis elevated), −1 if below the negative threshold (basis depressed), and 0 otherwise:

$$S1_t = \begin{cases} +1 & \text{if } z_t > \theta \\ -1 & \text{if } z_t < -\theta \\ 0 & \text{otherwise} \end{cases}$$

S2: Volatility Filter A regime filter based on the volatility ratio. Returns 1 (high volatility warning) if the ratio of short-term to long-term realized volatility exceeds its 90th percentile, and 0 otherwise:

$$S2_t = \mathbf{1}\{\text{vol_ratio}_t > \text{Percentile}_{90}\}$$

S3: Funding Rate HMM Regime Detection The HMM classifies the market into three regimes based on funding rate dynamics: *crowd long*, *crowd short*, and *neutral*. The signal is:

$$S3_t = \begin{cases} +1 & \text{crowd short (long signal)} \\ -1 & \text{crowd long (short signal)} \\ 0 & \text{neutral} \end{cases}$$

Trading Rules

The primary directional thesis is to go **short perp / long spot** when the market is crowded long, capturing basis convergence as the funding rate-driven premium reverts.

Entry Condition

A trade is entered when all three signals align:

$$\text{Entry} \iff S3 = 1 \text{ AND } S1 = 1 \text{ AND } S2 = 0$$

That is: the crowd is short (contrarian long signal from S3), the basis is elevated above threshold (S1), and volatility is not in an extreme regime (S2 = 0).

Exit Condition

A position is exited when:

$$\text{Exit} \iff S2 = 1 \text{ OR Funding Rate Flips OR Regime Changes}$$

Position sizing rule: If S2 = 1 (high volatility), reduce position size.

Funding Rate Analysis – HMM

Role

The HMM acts as a **sentiment oscillator**, identifying periods of crowded positioning in the perpetual futures market. Elevated positive funding indicates excessive long positioning, which historically precedes basis compression as longs are either liquidated or unwind voluntarily.

Model Features

The HMM is fitted on the following feature set (implemented in `hmm_features.py`):

- Basis change.
- Funding rate percentage change.
- Funding MA diff: 8-hour MA minus 24-hour MA of annualized funding.
- Realized volatility: 24-hour rolling annualized volatility.
- Perp Normalized Order Flow Imbalance (NOFI).
- Flow divergence: Perp NOFI minus Spot NOFI.

Model Specification

The HMM is implemented in `hmm.py` using `hmmlearn`'s `GaussianHMM`. The model is trained on the rolling 8-month window and applied forward to the 1-month evaluation period in the walk-forward framework. Three latent states are identified:

- **Crowd Long:** High funding, positive basis, bullish order flow.
- **Crowd Short:** Negative funding, negative basis, bearish order flow.
- **Neutral:** Low absolute funding, balanced flow.

The model outputs regime labels, regime probabilities, and trading signals. The probabilities are used to construct the discrete S3 signal, requiring a minimum 60% confidence before a regime is acted upon.

Implementation

`strategy.py` – Signal Generation

All three signals are implemented as standalone functions:

Signal	Function	Description
S1	`compute_S1()`	Z-score dislocation (mean-reversion); triggers at ± 1.5 threshold
S2	`compute_S2()`	Volatility filter; flags when `vol_ratio` exceeds its 90th percentile
S3	`compute_S3()`	HMM regime signal (fade the crowd); requires 60% confidence

Entry and exit logic is encapsulated in three additional functions:

- `compute_entry_signal()`: Entry when $S1 \neq 0$, $S2 = 0$, $S3 \neq 0$, and $\text{sign}(S1) = \text{sign}(S3)$.
- `compute_exit_signal()`: Exit on regime change, z-score zero-cross, or volatility spike.
- `signals_to_positions()`: Converts signal array to positions with carry-forward.

`backtest.py` – Walk-Forward Backtesting

The backtest integrates all components:

1. Signal functions are imported from `strategy.py` (lines 20–28).
2. `_run_single_fold()` (lines 390–412): Computes S1, S2, S3, and entry signals per test fold.
3. `_compute_basis_pnl()` (lines 486–622): Full PnL accounting covering spot leg, perp leg, funding PnL (at 8-hour intervals), and transaction costs.

Position convention: $d = +1$ denotes long spot / short perp (fade crowded long); $d = -1$ denotes short spot / long perp (fade crowded short). Strategy and cost parameters are configurable via `StrategyParams` (z-score and confidence thresholds) and `CostParams` (fees, slippage).